Indian National AI Olympiad - 2026 Stage 1 Test

Jan 18, 2026

Section A1

Cracking the Imperial Glyphs [3 pts]

Naruto, a time-travelling linguist, lands in an ancient palace, where people communicate in a mysterious language. In this language, the order of symbols matters and there may not be a one-to-one correspondence with English words. He begins learning the language by matching illustrated pictures with their meanings.

English	**Imperial Glyphs**
This is a beautiful apple.	這美蘋果
This is an apple on a tree.	這蘋果樹
They serve green apples in the palace.	服綠蘋果宮
Does the tree make the apple beautiful?	樹造蘋果美？
There is a palace on the black mountain.	這黑山宮
The mountain paints the palace black.	山塗宮黑

Output

Naruto now wants to compliment the palace gardener by correctly saying:

"The tree makes the palace beautiful."

Which imperial glyph message should he use?

Options:

A. 樹 造 宮 美	**B.** 宮 造 樹 美
C. 樹 美 造 宮	**D.** 美 樹 造 宮
E. 樹 造 美 宮	**F.** 宮 美 造 樹

Filtering Fakes with Faulty Oracle [3 pts]

You are building a fake-news classifier that operates on a pair consisting of a news headline and an accompanying image. You plan to use a small offline pretrained multimodal model (e.g., GPT-4o-mini, no internet access) as oracle to extract features, and train a supervised classifier on top of these features using provided true labels (Fake vs. True).

The feature-extraction prompt produces the following table.

ID	Feature Name	Prompt to the Model
F_1	Character count	"Count the number of characters in the headline."
F_2	Estimated publication year	"Guess the publication year of this headline."
F_3	Tone	"Classify the headline as sensational or neutral."
F_4	Image–headline match score	"Rate from 1–5 how well the image supports the headline."
F_5	Language identification	"Identify the language of the headline and return the ISO-639-1 two-letter code only (e.g., en,). If uncertain, return unk"
F_6	Named outlet prediction	"Which news outlet likely published this headline?"
F_7	Topic classification	"Classify topic: science, sports, politics, literature, finance, other."
F_8	Quote/attribution presence	"Does the headline contain a quote or attribution? (yes/no)"

Output

Which of the following feature subsets are structurally the most unreliable when produced by a small offline LLM and should be excluded?

Options:

A. $\{F_3, F_4, F_7\}$

B. $\{F_1, F_2, F_6\}$

C. $\{F_1, F_2, F_8\}$

D. $\{F_2, F_3, F_7\}$

E. $\{F_3, F_4, F_6\}$

F. $\{F_4, F_5, F_7\}$

Smart Student Search [3 pts]

You are a student intern helping to improve your school's student directory search system. Each student ID has the format: `(Section, FirstName, LastName)`, e.g., `(9C, Aryan, Shah)`. You plan to create a custom name similarity function by building on two standard string comparison measures:

- **Edit Distance:**

  $\mathrm{ED}(x, y)$ = minimum number of single-character insertions, deletions, or substitutions to transform x into y

- **Jaro Distance:**

$$\mathrm{JD}(x, y) = \begin{cases} \frac{1}{3}\left(\frac{m}{|x|} + \frac{m}{|y|} + \frac{m-t}{m}\right) & \text{if } m > 0 \\ 0 & \text{if } m = 0 \end{cases}$$

  where:

  - m = number of matching characters (characters from x and y that are the same and within $\lfloor \max(|x|, |y|)/2 \rfloor - 1$ positions of each other)
  - t = half the number of matching characters that are out of order between x and y.

Design Rules:

Your custom similarity function $\mathrm{Sim}(s, t)$ must satisfy the following requirements:

1. $\mathrm{Sim}(s, t) = 1$ if and only if all fields match exactly.
2. $\mathrm{Sim}(s, t) = 0$ if section codes differ.
3. $\mathrm{Sim}(s, t) = \mathrm{Sim}(t, s)$, i.e., it is symmetric.
4. When sections match, first name mismatches reduce the score more than last name mismatches.
5. $\mathrm{Sim}(s, t) \in [0, 1]$ for all s, t.

Let each student ID be written as $s = (c, f, l)$ and $t = (c', f', l')$, where the three fields denote the section code, first name, last name in that order and let $\mathbb{1}_{x=x'}$ denote the indicator function that equals 1 when $x = x'$ and 0 otherwise.

Output

Which among the following functions is a good choice for the custom similarity function $Sim(s, t)$?

Options:

A. $\mathbb{1}_{c=c'} \cdot \dfrac{\mathrm{JD}(f, f') + \mathrm{JD}(l, l')}{2}$

B. $\mathrm{JD}(c, c') \cdot \exp\big(-(1 - \mathrm{JD}(f, f')) - 2\big(1 - \mathrm{JD}(l, l')\big)\big)$

C. $\mathbb{1}_{c=c'} \cdot \exp\big(-2\big(1 - \mathrm{JD}(f, f')\big) - \big(1 - \mathrm{JD}(l, l')\big)\big)$

D. $\mathbb{1}_{c=c'} \cdot e^{-\mathrm{ED}(f,f')} \cdot e^{-2\mathrm{ED}(l,l')}$

E. $\mathbb{1}_{c=c'} \cdot \dfrac{2|f'|\mathrm{JD}(f, f') + |l'|\mathrm{JD}(l, l')}{2|f'| + |l'|}$

F. $\mathbb{1}_{c=c'} \cdot \big(2\mathrm{JD}(f, f') + \mathrm{JD}(l, l')\big)$

Recursive V-67 Outbreak [3 pts]

A highly contagious virus, $V-67$, is spreading in a population. There are three groups of people:

- N_t: number of **normal infectious** individuals at the start of day t,
- M_t: number of **mutant infectious** individuals at the start of day t,
- R_t: number of **removed** individuals (recovered or immune), who do not infect others and cannot be infected again.

Each day proceeds in the following order:

1. **New infections.** Each normal infectious person infects 2 new people and each mutant infects 4 new people. The total number of new infections on day t is
$$X_t = 2N_t + 4M_t.$$
2. **Fate of the newly infected.** Of these X_t newly infected people:
$$\frac{1}{2}X_t \text{ become normal infectious,} \quad \frac{1}{4}X_t \text{ become mutant infectious,} \quad \frac{1}{4}X_t \text{ are immediately removed.}$$
3. **Recovery of existing infectious individuals.** At the end of the day, half of the individuals who were infectious (of both types) at the start of the day recover:
$$\frac{1}{2}N_t \to R, \qquad \frac{1}{2}M_t \to R.$$
Newly infected individuals do not recover on the same day they are infected.

Output

Given that the outbreak starts on Day 0 with $N_0 = 8, \quad M_0 = 8, \quad R_0 = 0$, what is the total number of currently infectious individuals at the start of Day 3, that is, compute $N_3 + M_3$?

Options:

A) 265

B) 275

C) 281

D) 295

E) 305

F) 321

Pikachu's Path to Victory [3 pts]

Pikachu is preparing for his ultimate challenge on a battlefield made of **Thunder Pads** arranged in a 5×5 grid as shown in Figure 1. Each pad crackles with electric energy that Pikachu can absorb on his journey to become victorious. He starts on the pad at the top-left $(0, 0)$ and must reach the final pad in the bottom-right corner $(4, 4)$.

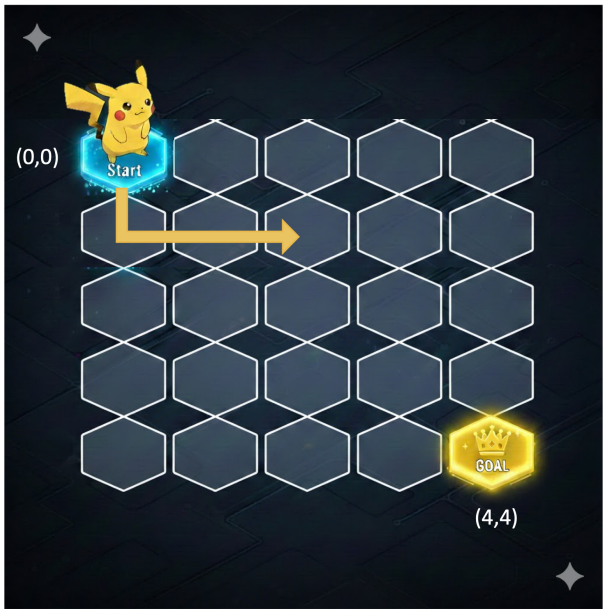


Figure 1: Pikachu's Thunder Pad battlefield.

Rules of Movement:

- **Thunder Leap:** Pikachu moves exactly like a knight in chess: two squares in one cardinal direction and one square perpendicular to it.

- **Spark Charge:** Every time Pikachu lands on a pad, he *activates* it and absorbs its energy.
- **One-Time Power:** Once a pad has been activated, Pikachu *cannot* land on it again.

Output

What is the **maximum number of pads** Pikachu can activate (including both the start and the final pad) on a valid path from $(0, 0)$ to $(4, 4)$?

Options:

A) 15

B) 19

C) 20

D) 21

E) 25

F) 28

Section A2

The Phulera Godown Showdown [3 pts]

In the village of Phulera, the four biggest rice fields are denoted as $R_i(x_i, y_i)$ for $i = 1, 2, 3, 4$, and are located at the coordinates as shown in Figure 2:

$$R_1(40, 180), \quad R_2(120, 80), \quad R_3(200, 140), \quad R_4(80, 20).$$

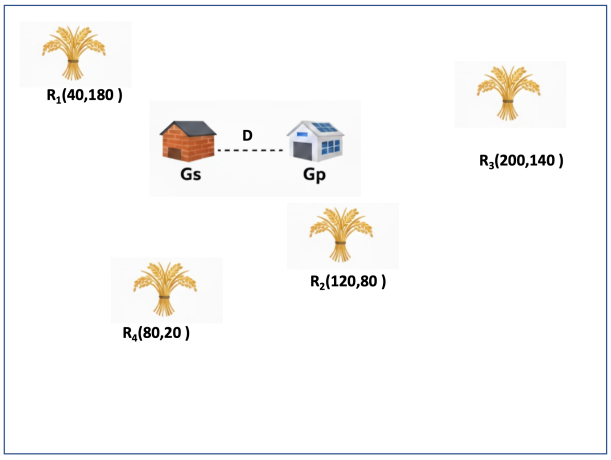


Figure 2: Phulera's rice fields and godown placement. Not to scale.

Sachivji argues that because the village has criss-cross perpendicular roads, travel cost is best modeled using Manhattan (L1) distance. He proposes placing the village godown at

$$G_S = (x_S, y_S) = \arg\min_{(x,y)} \sum_{i=1}^{4} \left(|x - x_i| + |y - y_i| \right).$$

Pradhanji, to showcase his advanced thinking, proposes modeling using an information divergence and proposes placing the godown at

$$G_P = (x_P, y_P) = \arg\min_{(u,v)} \sum_{i=1}^{4} \left[x_i \ln \frac{x_i}{u} + y_i \ln \frac{y_i}{v} + (u - x_i) + (v - y_i) \right].$$

Output

What is $\lfloor D \rfloor$ if D is the Euclidean distance between Pradhanji and Sachivji's godowns, i.e.,

$$D = \sqrt{(x_S - x_P)^2 + (y_S - y_P)^2}.$$

Note: If multiple points satisfy Pradhanji or Sachivji's conditions, they choose the arithmetic mean of the possible coordinate ranges to determine xs and ys.

The Blind Board Gambit [3 pts]

In a 3-board team chess match, Team Gukesh and Team Anand each have three players. Each team secretly and independently assigns one player to each board, and players on the same board compete against each other. Each game is played until one player wins (no draws). A team wins the match if it wins at least 2 of the 3 board games. Results across boards are independent.

Ratings and Win Probability: When a player with rating R plays against a player with rating S, the probability that the R-rated player wins is

$$P(R \text{ beats } S) = \frac{1}{1 + 10^{(S-R)/400}}.$$

The teams have the following ratings:

- Team Gukesh: $A = 2000$, $B = 1800$, $C = 1600$
- Team Anand: $X = 1900$, $Y = 1700$, $Z = 1500$

Scenario: A spy tips off Anand that Gukesh will **NOT** use the canonical assignment (A, B, C). Believing Gukesh will choose uniformly from the 5 remaining orderings, Anand selects the assignment of (X, Y, Z) that maximizes his expected win probability. Gukesh independently assigns (C, A, B) to Boards $(1, 2, 3)$ in accordance with the tip.

Output

Let p be the probability that Team Anand wins the match under this scenario. Compute $\lfloor 100p \rfloor$.

The Crude Curve Forecast [3 pts]

Prof. Clairvoyant, an oil market analyst, models crude prices by decomposing them into multiple effects: a baseline level, long-term trends, seasonal fluctuations, geopolitical shocks, and market corrections. Using observations of the monthly average crude oil price per barrel $P(t)$, he arrives at the following continuous time model:

$$P(t) = A + Bt + C \cdot (1 - e^{-(t-\tau_C)}) \cdot h(t - \tau_C) + D \cdot h(t - \tau_D) + E \cdot \sin\left(\frac{2\pi t}{\tau_E}\right)$$

where t is time in months from Jan 2020 ($t = 0$), the parameters A, B, C, D, E, T, τ_A, τ_B, τ_D are all unknown, and $h(t)$ is the Heaviside step function defined as

$$h(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}.$$

Figure 3 shows $P(t)$ for $0 \leq t \leq 60$ months. The graph is drawn to scale with key values readable at strategic time points.

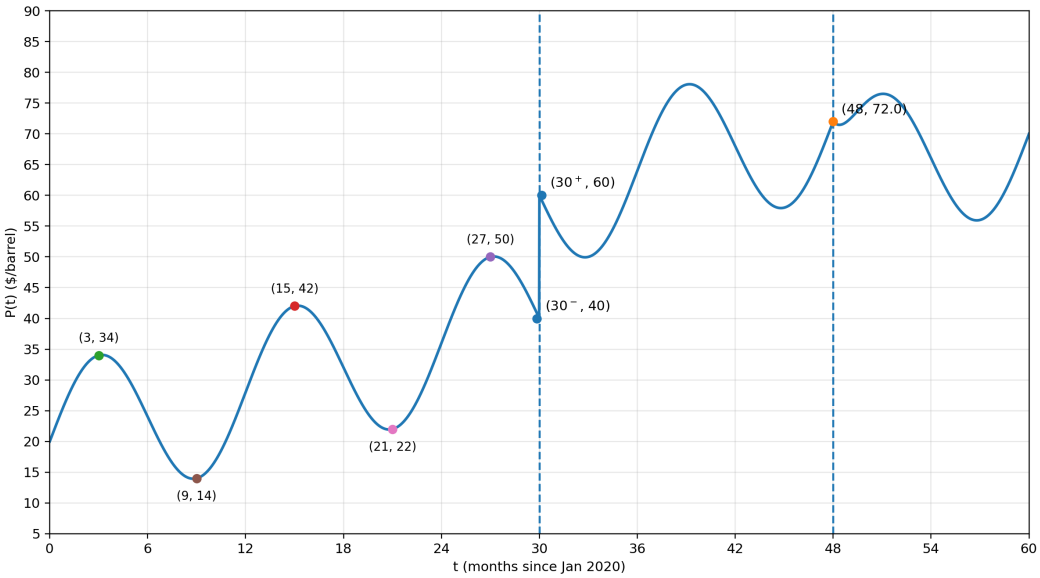


Figure 3: Monthly crude oil price variation with time.

Output

Using the price model and the above graph, estimate $\lfloor P(75) \rfloor$ where $P(75)$ is the price at $t = 75$ months.

The Catalyst Constraint [3 pts]

You are a pharmaceutical process engineer optimizing the production of drug compound **D**. Two parallel synthesis pathways can produce D:

$$\textbf{Pathway A:} \quad \text{A} \xrightarrow{C} \text{D}$$
$$\textbf{Pathway B:} \quad \text{B} \xrightarrow{C} \text{D}$$

where A and B are different substrate molecules, and C is a precious catalyst that must be shared between both pathways.

Reaction Kinetics: The yield of compound D from each pathway follows quadratic kinetics due to catalyst inhibition effects:

$$\textbf{Pathway A: } d_A = a \cdot c_A - 2c_A^2, \qquad \textbf{Pathway B: } d_B = b \cdot c_B - 3c_B^2,$$

where a, b are moles of substrates A and B available; c_A, c_B are moles of catalyst C allocated to each pathway; and d_A, d_B are moles of product D from each pathway.

Output

Your production batch has the following resources: 16 moles of substrate A, 24 moles of substrate B, and 5 moles of catalyst C.

What is the maximum total yield of compound D (in moles) that you can produce by optimally allocating catalyst C between the two pathways? Report $\lfloor d_A + d_B \rfloor$ rounded down to the nearest integer.

The Pit-Stop Paradox [3 pts]

It is the final 10 laps of the Monaco Grand Prix. Max Verstappen (Driver V) is leading Lewis Hamilton (Driver H), but Verstappen's tires are critically worn while Hamilton's are in better condition. Both teams must decide: **Pit for fresh tires or stay out**.

Strategy Probabilities (Weather-Dependent): Each driver independently chooses whether to pit in the final 10 laps. Historical data and team behavior suggest the following probabilities:

Strategy	**P(Strategy | No Rain)**	**P(Strategy | Rain)**
Neither pits	0.45	0.05
Only V pits	0.30	0.10
Only H pits	0.10	0.10
Both pit	0.15	0.75

Race Outcome Probabilities: Race simulations estimate the probability that Hamilton wins under each scenario:

Strategy	**P(H wins | Strategy, No Rain)**	**P(H wins | Strategy, Rain)**
Neither pits	0.25	0.50
Only V pits	0.80	0.05
Only H pits	0.10	0.95
Both pit	0.55	0.60

Output

You arrive after the checkered flag and learn that Hamilton won the race. You do not know whether it rained or which pit strategies were chosen.

Given that there was a prior 50% chance of rain based on the weather forecast, what is your estimate of the probability that at least one driver pitted? Report $\lfloor 100 \times P(\text{at least one pitted} \mid H \text{ wins}) \rfloor$.

Section B1

Code Triage: Critical Classification Challenge

Background: Multi-Class Classification

In a multi-class classification problem, the goal is to assign instances to one of several possible classes. The performance of the classifier is evaluated using the following metrics.

Confusion Matrix: The confusion matrix for a 3-class problem is structured as:

$$\begin{bmatrix} \text{TP}_{C_1} & F_{C_1,C_2} & F_{C_1,C_3} \\ F_{C_2,C_1} & \text{TP}_{C_2} & F_{C_2,C_3} \\ F_{C_3,C_1} & F_{C_3,C_2} & \text{TP}_{C_3} \end{bmatrix}$$

where:

- TP_{C_i}: True positives for class C_i (instances correctly classified as C_i)
- F_{C_i,C_j}: Instances of class C_j incorrectly classified as C_i

Binary Classification Metrics: For multi-class problems, we can treat one class as "positive" and all others as "negative" to compute binary metrics.
Sensitivity (True Positive Rate): For class C_i as positive, sensitivity measures the proportion of actual C_i instances correctly identified:

$$\text{Sensitivity}_i = \frac{\text{TP}_{C_i}}{\text{TP}_{C_i} + \sum_{j \neq i} F_{C_j,C_i}}$$

Specificity (True Negative Rate): For class C_i as positive, specificity measures the proportion of actual non-C_i instances correctly identified as non-C_i:

$$\text{Specificity}_i = \frac{\text{TN}_{C_i}}{\text{TN}_{C_i} + \sum_{j \neq i} F_{C_i,C_j}}$$

where TN_{C_i} is the number of instances correctly classified as not C_i.

ROC Curve: The Receiver Operating Characteristic (ROC) curve plots True Positive Rate (TPR = Sensitivity) against False Positive Rate (FPR = 1 - Specificity) at various decision thresholds.

Problem Context

A hospital emergency department uses an automated triage system based on a linear classifier that analyzes patient vital signs, lab results, and symptoms to output a **risk score**. Based on this score and two thresholds (t_1 and t_2), patients are classified into three ordered severity levels:

- **Class 1 - Outpatient:** Score $< t_1$, low severity
- **Class 2 - Inpatient Regular Ward:** $t_1 \leq$ Score $< t_2$, moderate severity
- **Class 3 - Inpatient ICU:** Score $\geq t_2$, high severity

This classifier was tested on 1000 patients with the following actual distribution:

- Outpatient care (Class 1): 60%
- Regular Ward (Class 2): 30%

- ICU care (Class 3): 10%

The classifier achieves the following performance:

Class	Sensitivity	Specificity
Class 1 (Outpatient)	0.85	0.90
Class 2 (Regular Ward)	0.70	0.89
Class 3 (ICU)	0.80	0.91

Analysis of the score distributions shows that egregious errors are low:

- 3.33% of actual Class 1 patients are misclassified as Class 3.
- 10% of actual Class 3 patients are misclassified as Class 1.

Questions:

Q1: Missed ICU Patients [1 pt]

Output

What is the fraction x of actual ICU patients (Class 3) who were incorrectly classified as non-ICU (either Class 1 or Class 2). Report your answer as percentage rounded down to nearest integer $\lfloor 100x \rfloor$.

Q2: Outpatient Triaging [1 pt]

Output

Among the patients who actually need only outpatient care (Class 1). what fraction (x) does the system correctly identify as not requiring hospitalization (i.e., correctly classified as Class 1 rather than Class 2 or Class 3)? Report your answer as percentage rounded down to nearest integer $\lfloor 100x \rfloor$.

Q3: Operating Point for ICU Detection [2 pts]

The existing triage system can be reconfigured without any classifier retraining by adjusting the score thresholds. For ICU detection (Class 3 as positive vs. Classes 1 and 2 as negative), varying the threshold t_2 produces the ROC curve shown in Figure 4 with different possible operating points.

The hospital serves on average 200 patients per day. Since the ICU has only 25 beds, the system must be configured such that the expected number of patients flagged for ICU does not exceed 25 per day.

Output

Which operating point maximizes sensitivity while respecting the capacity constraint of 25 beds?

Options:

1. Point A (FPR = 0.02, TPR = 0.60).

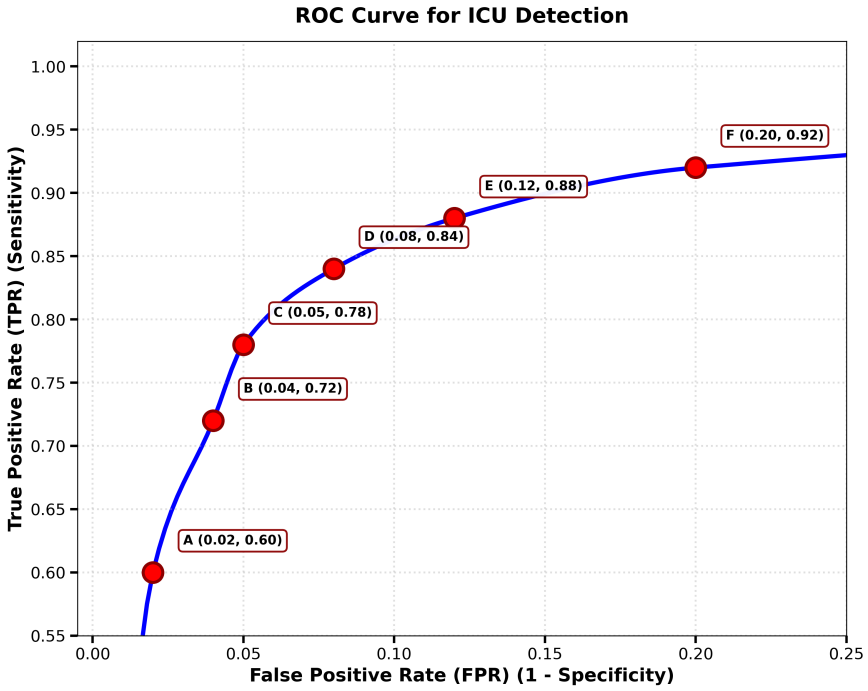


Figure 4: ROC curve for ICU detection showing six operating points (A through F).

2. Point B (FPR = 0.04, TPR = 0.72).
3. Point C (FPR = 0.05, TPR = 0.78).
4. Point D (FPR = 0.08, TPR = 0.84).
5. Point E (FPR = 0.12, TPR = 0.88).
6. Point F (FPR = 0.20, TPR = 0.92).

Q4: ROC Curve - Effect of Score Transformation [1 pt]

One of the experts suggests applying a monotonic transformation to the risk score:

$$s_{\text{new}} = s_{\text{old}}^{1.5},$$

where $s_{\text{old}} \in \mathbb{R}$ is the original score.

Output

How does this transformation affect the ROC curve for ICU detection (Class 3 vs. others) assuming nothing else changes ?

Options:

1. The curve improves consistently (toward top-left).
2. The curve worsens consistently (away from top-left).

3. The curve remains unchanged.

4. The curve improves only if the transformation increases the gap between Class 3 and Class 2 means.

5. The curve improves at higher FPR values (right side, lower specificity).

6. The curve improves at lower FPR values (left side, higher specificity).

Q5: ROC Curve - Effect of Distribution Shift [1 pt]

Consider the ROC curve from Q3 for ICU detection. Suppose the patient population changes such that the proportion requiring ICU increases from 10% to 25%, while the classifier's scoring function and decision threshold remain unchanged.

Output

How does this change affect the ROC curve for ICU detection (Class 3 vs. others) assuming everything else remains the same ?

Options:

1. The curve shifts uniformly upward (toward top-left).

2. The curve shifts uniformly downward (away from top-left).

3. The curve remains unchanged.

4. The left part (low FPR) shifts upward while the right part (high FPR) shifts downward.

5. The left part (low FPR) shifts downward while the right part (high FPR) shifts upward.

6. The curve compresses horizontally.

Q6: Designing an Ordinal Classifier System [2 pts]

You are the chief scientist for the hospital. Medical domain analysis shows that criteria for distinguishing Outpatient vs. Inpatient (Classes 2 and 3) differ from the criteria for Regular Ward vs. ICU (e.g., pain/infection markers vs. vital instability/organ function). You wish to design a new advanced classification system for the three-class triage problem but prefer to stick with linear models so that the decisions are interpretable for doctors and training data is limited.

You are considering the following architectures and loss function:

Architectures:

- **X:** Single linear classifier with two thresholds ($t_1 < t_2$). (as described earlier)
- **Y:** Two binary classifiers: (Class 1 vs. Classes {2,3}), (Classes {1,2} vs. Class 3)
- **Z:** Three pairwise classifiers: (Class 1 vs. Class 2), (Class 2 vs. Class 3), (Class 3 vs. Class 1).

Loss Functions: Let y be the true class label $\in \{-1, 1\}$ and $f(x)$ the predicted score:

- **Squared Loss [L_{SQ}]:** $(y - f(x))^2$

- **Hinge Loss** $[L_{HG}]$ **:** $\max(0, 1 - y \cdot f(x))$
- **Logistic Loss** $[L_{LOG}]$ **:** $\log(1 + e^{-y \cdot f(x)})$

Consider the following claims:

- **Claim 1:** **Y** and **Z** may produce inconsistent predictions in terms of case severity requiring post-processing.
- **Claim 2:** **Y** has a high chance of achieving better accuracy than **X** because it can model the class boundaries more accurately.
- **Claim 3:** **Z** has a high chance of achieving better accuracy than **Y** because it can model the class boundaries more accurately.
- **Claim 4:** For **X**, L_{SQ} is better suited than L_{HG} at enforcing ordinal structure, but poor at boundary estimation in the presence of outliers.

  Claim 5: For **Y**, L_{SQ} is more robust to outliers than L_{LOG} because L_{SQ} provides smooth gradients throughout the entire range.

Output

Which combination of claims is correct?

Options:

1. Claims 1, 2, 3, and 4 only
2. Claims 1, 2, 4, and 5 only
3. Claims 1, 2, 3, and 5 only
4. Claims 1, 2, and 4 only
5. Claims 2, 3, and 4 only
6. Claims 1, 3, and 4 only

Q7: Triage Benefits [2 pts]

The hospital assigns benefit scores (in quality points) for each classification decision:

	Predicted Class 1	**Predicted Class 2**	**Predicted Class 3**
Actual Class 1	100	50	0
Actual Class 2	0	100	50
Actual Class 3	0	50	200

Output

For the simple triage classifier described earlier (single linear model with two thresholds) with sensitivity, specificity, and error distribution as in the original context, use the benefit matrix above to calculate the expected benefit (x) per patient averaged over all 1000 patients. Report your answer as an integer $\lfloor x \rfloor$.

Section B2

WonkaRank: The Golden Ticket Retrieval Challenge

Background: Information Retrieval Metrics

In information retrieval, documents are represented as vectors in a high-dimensional space and ranked by a score function to retrieve documents relevant to a user's need.

Vector Representation: Given a vocabulary $V = \{t_1, t_2, \ldots, t_m\}$ of m terms, each document $D^{(j)}$ is a vector in $\mathbb{R}^m$:

$$D^{(j)} = (d_1^{(j)}, d_2^{(j)}, \ldots, d_m^{(j)})$$

where $d_i^{(j)}$ is the weight of term t_i in document $D^{(j)}$.

Relevance: Each document has a true relevance $\text{Rel}(D^{(j)}) \in \mathbb{R}^+$, where higher values indicate greater relevance. A document is relevant if $\text{Rel}(D^{(j)}) > 0$. Ties are broken by numeric Chocolate ID in ascending order ($D^{(1)} < D^{(2)} < \ < D^{(10)}$).
Ranking Function: A score function $\text{Score}(D^{(j)}) \in \mathbb{R}$ assigns a score to each document. Documents are ranked in decreasing score order: $D^{(1)}, D^{(2)}, D^{(3)}, \ldots$

Retrieval Metrics:
(a) Precision and Recall: Given the top k results and R total relevant documents:

$$\text{Recall@}k = \frac{\text{Number of relevant documents in top } k \text{ results}}{R}$$

$$\text{Precision@}k = \frac{\text{Number of relevant documents in top } k \text{ results}}{k}$$

where a document is relevant if $\text{Rel}(D) > 0$.
(b) Discounted Cumulative Gain (DCG): DCG measures ranking quality by considering both relevance values and document positions:

$$\text{DCG@}k = \sum_{i=1}^{k} \frac{\text{Rel}(D^{(i)})}{\log_2(i+1)}$$

where $D^{(i)}$ is the document at position i. The logarithmic discount emphasizes placing highly relevant documents early. Here, @k indicates that the DCG is computed considering only the top k ranked documents.
(c) Position-Based Inspection (PI-Recall@k): Users often do not examine all results due to attention or satisfaction decay. Let p_i be the probability that a user inspects position i. Position-based Inspection Recall is:

$$\text{PI-Recall@}k = \frac{1}{R} \times \sum_{i=1}^{k} p_i \times \mathbb{1}(\text{Rel}(D^{(i)}) > 0)$$

where $(\cdot)$ is the indicator function (1 if true, 0 if false).

Problem Context

Willy Wonka has hidden Golden Chocolate Factory Tour tickets inside a small number of premium chocolates sold across the city to celebrate the New Year. Each chocolate lists the quantities of four ingredients on its wrapper: Cocoa (C), Hazelnut (H), Caramel (K), and Mint (M).

Charlie wants to find tour tickets while minimizing spending, so he designs ranking strategies that place chocolates with tickets early in his purchase order.

Celebration Chocolate Collection

The city has 9 premium chocolates with the following ingredients [Cocoa (C), Hazelnut (H), Caramel (K), and Mint (M)] and ticket information.

Chocolate ID	C	H	K	M	Golden Ticket?	Relevance
D_1	1	1	0	0	VIP Tour	10
D_2	1	1	1	0	No Ticket	0
D_3	1	0	1	0	No Ticket	0
D_4	0	1	0	0	No Ticket	0
D_5	1	1	0	1	Partial Tour	3
D_6	0	0	1	1	No Ticket	0
D_7	1	0	0	0	No Ticket	0
D_8	1	1	0	0	Full Tour	5
D_9	0	1	1	0	Partial Tour	3

A chocolate is considered relevant if it contains any tour ticket (Relevance > 0).

Questions:

Q1: Recall@k Calculation [1 pt]

Charlie purchases the following 6 chocolates in order: $[D_2, D_3, D_5, D_6, D_7, D_8]$.

Output

What is the recall percentage for top-5 results, i.e., $\lfloor 100 \times \text{Recall@5} \rfloor$?

Q2: Comparison of Ranking Strategies [1 pts]

Charlie compares two scoring-based ranking strategies. Both rank chocolates by decreasing score with ties broken arbitrarily.

Strategy A:

$$\text{Score}_A(D) = \begin{cases} 2 & \text{if chocolate contains both Cocoa and Hazelnut} \\ 0 & \text{otherwise} \end{cases}$$

Strategy B:

$$\text{Score}_B(D) = \begin{cases} 2 & \text{if chocolate contains both Cocoa and Hazelnut} \\ 1 & \text{if chocolate contains exactly one of Cocoa or Hazelnut} \\ 0 & \text{if chocolate contains neither} \end{cases}$$

Output

Which statement is correct about Strategy B compared to Strategy A? (Ties are broken by numeric Chocolate ID in ascending order.)

Options:

1. Strategy B improves recall but reduces precision.
2. Strategy B improves precision but reduces recall.
3. Strategy B maintains recall but reduces precision.
4. Strategy B improves both precision and recall.
5. Both strategies produce identical recall.
6. None of the above.

Q3: Discounted Cumulative Gain [2 pts]

Charlie evaluates ranked lists using the discounted cumulative gain metric (DCG) defined earlier. He tries the following two strategies that return the same 4 chocolates but in different orders:

Strategy A: $[D_5, D_1, D_8, D_2]$
Strategy B: $[D_8, D_5, D_1, D_2]$

Output

Which property of the DCG metric would result in preference for Strategy B over Strategy A?

Options:

1. The metric depends only on how many relevant chocolates appear, not their order.
2. The metric gives strong weight to placing the highest-relevance document as early as possible.
3. The metric treats all non-zero relevance values as equal but heavily discounts later ranks.
4. The metric ignores all ranks after the first position.
5. The metric penalizes irrelevant chocolates more than it rewards relevant ones.
6. None of the above.

Q4: Optimal Scoring Function [3 pts]

Let each chocolate D have ingredient vector (d_C, d_H, d_K, d_M) and let $W = (w_C, w_H, w_K, w_M) = (3, 3, 1, 1)$ denote the ingredient weight vector.

Charlie considers four scoring functions where $i \in \{C, H, K, M\}$:

- $\textbf{Score}_A(D) = \sum_i d_i \cdot w_i$
- $\textbf{Score}_B(D) = \sum_i \min(1, d_i) \cdot \min(1, w_i)$
- $\textbf{Score}_C(D) = (\sum_i d_i \cdot w_i) - (\sum_i d_i)$
- $\textbf{Score}_D(D) = \dfrac{\sum_i d_i \cdot w_i}{1 + \sum_i d_i}$

Output

Which scoring function(s) achieve the highest Precision@2 when retrieving the top 2 results?(Ties are broken by numeric Chocolate ID in ascending order.)

Options:

1. Score_A only.
2. Score_B only.
3. Score_C only.
4. Score_D only.
5. All four achieve the same Precision@2.
6. Multiple (but not all) achieve the highest Precision@2.

Q5: Position-based Inspection Effects [3 pts]

Charlie uses Score_A with weights $W = (3, 3, 1, 1)$ to produce a chocolate ranking. His probability of buying the chocolates (i.e., inspection) in the ranked order is given by:

$$p = [p_1, p_2, p_3, p_4, p_5] = [0.5, 0.5, 0.3, 0.3, 0.2],$$

where p_i is the probability that he buys i^{th} ranked chocolate.

Output

Compute the expected recall percentage $\lfloor 100 \times \text{PI-Recall@5} \rfloor$ under this inspection model. (Ties are broken by numeric Chocolate ID in ascending order.)

Section B3

Pattern Pursuit: Convolution in Signal Processing

Background: Time Series and Convolution

Time Series: A time series is an ordered sequence of numerical values measured over time, appearing in applications like weather forecasting, financial markets, health tracking, and sensor data. We denote a time series as $\mathbf{x} = [x_1, x_2, x_3, \ldots, x_n]$, where n is the length.

Kernel: A kernel is a short sequence $\mathbf{h} = [h_1, h_2, \ldots, h_k]$ used to detect patterns in a time series. For example, $\mathbf{h} = [1, -1]$ detects changes between consecutive values.

Convolution: Convolution applies a kernel to a time series to extract features. The output is computed as:

$$y_i = \sum_{j=1}^{k} h_j \cdot x_{i+k-j}$$

where the output length is $m = n - k + 1$ (when no padding is used).
The **aggregate outputs** S and S^{abs} are defined in terms of all convolution values:

$$S = \sum_{i=1}^{m} y_i; \quad S^{abs} = \sum_{i=1}^{m} |y_i|$$

Computing Convolution (3 steps):

1. Reverse the kernel
2. Slide the reversed kernel along the time series
3. At each position, multiply corresponding elements and sum

Example: Given $\mathbf{x} = [2, 4, 6, 8]$ and $\mathbf{h} = [1, -1]$:

Reversed kernel:	$[-1, 1]$
Position 1:	$(-1)(2) + (1)(4) = 2$
Position 2:	$(-1)(4) + (1)(6) = 2$
Position 3:	$(-1)(6) + (1)(8) = 2$
Output:	$\mathbf{y} = [2, 2, 2]$
Aggregates:	$S = S^{abs} = 2 + 2 + 2 = 6$

Padding: Adding values (typically zeros) to the ends of $\mathbf{x}$ before convolution. For example, padding $[3, 5, 7]$ with one zero on each side (i.e padding of 1) gives $[0, 3, 5, 7, 0]$. Unless stated otherwise, assume **no padding**.

Stride: The step size when sliding the kernel. Stride = 1 (default) means moving one position at a time; stride = 2 means skipping every other position. Unless stated otherwise, assume **stride = 1**.

Questions

Q1: Convolution with Custom Kernel [1 pt]

You are given the following time series and kernel:

$$\mathbf{x} = [2, 1, 3, 4], \quad \mathbf{h} = [1, 2]$$

Task: Apply convolution with the following parameters:

- Use padding of 1 with value 3 on both ends (prepend and append one copy of the value 3)
- Use stride = 2

Output

Select the correct convolution output $\mathbf{y}$ from the options below.

Options:

A. $[8, 5, 11]$

B. $[7, 4, 10]$

C. $[5, 7, 9]$

D. $[8, 11]$

E. $[11, 8]$

F. $[10, 7]$

Q2: Signal Smoothness vs. Fluctuations [2 pts]

Consider two time series with the same minimum (2) and maximum (10) values:

$$\mathbf{x} = [2, 4, 6, 8, 10], \quad \mathbf{z} = [2, 6, 3, 10, 4]$$

Signal $\mathbf{x}$ is smooth (values increase steadily), while signal $\mathbf{z}$ fluctuates (values jump up and down). We wish to find a kernel that distinguishes between these two types of signals.

Task: Find kernel $\mathbf{h} = [h_1, h_2, h_3]$ such that the aggregate output $S = \sum_{i=1}^{m} y_i$, assuming stride = 1, no padding, satisfies:

- Constant signals $\rightarrow S = 0$
- Smooth signals like $\mathbf{x} \rightarrow |S|$ small
- Fluctuating signals like $\mathbf{z} \rightarrow |S|$ large

Output

Select the kernel that best detects signal volatility.

Options:

A. $[1, 1, 1]$

B. $[-1, 0, 1]$

C. $[1, -2, 1]$

D. $[-1, 1, 0]$

E. $[0, 1, 0]$

F. $[1, 0, -1]$

Q3: Signal Acceleration [2 pts]

In time series analysis, **acceleration** measures whether a signal is speeding up or slowing down. Consider two time series:

$$\mathbf{x} = [1, 2, 4, 7, 11], \quad \mathbf{w} = [1, 5, 8, 10, 11]$$

Signal **x accelerates**, i.e., differences between consecutive values increase ($1 \to 2 \to 3 \to 4$) while signal **w decelerates**, i.e., differences between consecutive values decrease ($4 \to 3 \to 2 \to 1$).

Task: Find kernel $\mathbf{h} = [h_1, h_2, h_3]$ such that the aggregate output $S = \sum_{i=1}^{m} y_i$ (stride = 1, no padding) satisfies:

- Signals with constant velocity (constant differences between consecutive values) $\to S = 0$
- Accelerating signals like $\mathbf{x} \to S > 0$
- Decelerating signals like $\mathbf{w} \to S < 0$

Output

Select the kernel that best detects signal acceleration.

Options:

A. $[1, 1, 1]$

B. $[-1, 0, 1]$

C. $[1, -2, 1]$

D. $[-1, 1, 0]$

E. $[1, 0, -1]$

F. $[0, 1, -1]$

Q4: Spikes in Signals [2 pts]

A **spike** is a value significantly larger than both neighbors. Consider two signals:

$$\mathbf{x} = [2, 3, 8, 3, 2, 3], \quad \mathbf{z} = [2, 3, 4, 5, 6, 7]$$

Signal **x** contains a spike (value 8 is higher than both its left and right neighbors), while signal **z** is smooth (no spikes).

Task: Find kernel $\mathbf{h} = [h_1, h_2, h_3]$ such that $S^{abs} = \sum_{i=1}^{m} |y_i|$ (stride = 1, no padding) satisfies:

- Constant signals $\to S^{abs} = 0$
- Smooth signals $\to S^{abs}$ small

- Signals with spikes $\rightarrow S^{abs}$ large

Output

Select the kernel that best detects spike presence.

Options:

A. $[1, 1, 1]$

B. $[-1, 2, -1]$

C. $[-1, 0, 1]$

D. $[0, 1, 0]$

E. $[1, -1, 1]$

F. $[0, 1, -1]$

Q5: Signal Symmetry [3 pts]

A signal is **symmetric** if it reads the same forwards and backwards. Consider the two time series:

$$\mathbf{x} = [2, 5, 5, 2], \quad \mathbf{z} = [1, 2, 3, -1].$$

Signal $\mathbf{x}$ is symmetric while $\mathbf{z}$ is not.

Task: For length-4 signals $\mathbf{x} = [x_1, x_2, x_3, x_4]$, find kernel $\mathbf{h} = [h_1, h_2, h_3, h_4]$ such that the convolution output y (stride = 1, no padding) satisfies:

- Symmetric signals $\rightarrow y = 0$
- Asymmetric signals $\rightarrow y \neq 0$

Output

Select the kernel that detects symmetry by outputting zero for symmetric patterns.

Options:

A. $[-1, 2, -1, 0]$

B. $[1, -1, -1, 1]$

C. $[1, -2, 1, 0]$

D. $[1, -1, 1, -1]$

E. $[1, -2, 0, -1]$

F. $[1, 0, -1, 0]$

Section B4

Model Compression: Building Efficient Classifiers

Background: Binary Classification with Linear Models

In binary classification, each example $\mathbf{x}_i \in \mathbb{R}^d$ is assigned a label $y_i \in \{0, 1\}$. A linear classifier computes a score:

$$z_i = \mathbf{w}^\top \mathbf{x}_i + b$$

where $\mathbf{w} \in \mathbb{R}^d$ is the **weight vector**, $b \in \mathbb{R}$ is the **bias term** (intercept), and the prediction rule is:

$$\hat{y}_i = \begin{cases} 1 & \text{if } z_i > 0 \\ 0 & \text{if } z_i \leq 0 \end{cases}$$

The decision boundary is the hyperplane $\{\mathbf{x} : \mathbf{w}^\top \mathbf{x} + b = 0\}$. Note that the classification threshold is at exactly 0, with strict inequality: scores > 0 predict class 1, while scores ≤ 0 predict class 0.

Matrix Notation: For a dataset with n examples, we write $\mathbf{s} = \mathbf{X}\mathbf{w} + b\mathbf{1}$, where $\mathbf{X} \in \mathbb{R}^{n \times d}$ is the feature matrix (each row is one example), $\mathbf{s} \in \mathbb{R}^n$ is the score vector, and $\mathbf{1} \in \mathbb{R}^n$ is the all-ones vector. When $b = 0$, we have $\mathbf{s} = \mathbf{X}\mathbf{w}$.

Accuracy: Measures the fraction of correctly classified examples:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{n} \times 100\%$$

Indexing Convention: We use 1-based indexing throughout this problem. Features are indexed from 1 to d, and examples are indexed from 1 to n.

Background: Feature Pruning and Bias Correction

Why Prune?
In many applications, we simplify models by removing features. This:

- Reduces computational cost
- Decreases memory requirements
- Can improve interpretability
- May prevent overfitting

Feature Pruning: Pruning a feature j means setting its weight to zero: $w_j \leftarrow 0$.
If we prune a set of features $S \subseteq \{1, \ldots, d\}$, the score changes from:

$$z_i = \mathbf{w}^\top \mathbf{x}_i + b \quad \text{to:} \quad z_i^{(\text{pruned})} = \sum_{k \notin S} w_k x_{ik} + b$$

The removed contribution is:

$$r_i^{(S)} = \sum_{j \in S} w_j x_{ij}$$

Therefore:

$$z_i^{(\text{pruned})} = z_i - r_i^{(S)}$$

Bias Correction: After pruning, the average score across the dataset may shift. A common technique is **bias correction**: adjust the bias so that the mean score is preserved.
The correction formula is:

$$b' = b + \mu_S$$

where $\mu_S = \frac{1}{n} \sum_{i=1}^{n} r_i^{(S)}$ is the average removed contribution.
The corrected score becomes:

$$z_i^{(\text{corrected})} = z_i^{(\text{pruned})} + \mu_S = z_i - r_i^{(S)} + \mu_S$$

This reduces systematic drift and can improve post-pruning accuracy without retraining.

Questions:

Q1: Reverse Engineering the Weights [4 pts]

You are given a binary classification task with **6 examples** and **4 features**. Let $\mathbf{X} \in \mathbb{R}^{6 \times 4}$ be the feature matrix and $\mathbf{w} \in \mathbb{R}^4$ be the weight vector (with no bias term, i.e., $b = 0$). The score vector is $\mathbf{s} = \mathbf{X}\mathbf{w} \in \mathbb{R}^6$, and predictions follow the rule above.

Dataset:

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

You are given two possible score vectors s^A and s^B. Each has **4 known values** and **2 unknown values** (marked with ?). All known score values are **sign-consistent** with $\mathbf{y}$ (i.e., positive when $y_i = 1$, negative when $y_i = 0$).

$$\mathbf{s}^{(A)} = \begin{bmatrix} 2 \\ ? \\ 1 \\ -1 \\ 1 \\ ? \end{bmatrix}, \quad \mathbf{s}^{(B)} = \begin{bmatrix} 3 \\ -5 \\ ? \\ -1 \\ ? \\ 1 \end{bmatrix}$$

Task:

1. Solve for the weight vector $\mathbf{w}$ from the equation $\mathbf{s} = \mathbf{X}\mathbf{w}$.
2. Use the solved weights to fill in the missing scores (?).
3. Compute predictions $\hat{\mathbf{y}}$ and calculate the accuracy.
4. Choose the option with higher accuracy.

Output

Let $\mathbf{w}^*$ be the weight vector from the option with higher accuracy. Compute the ℓ_1 magnitude:

$$\|\mathbf{w}^*\|_1 = |w_1^*| + |w_2^*| + |w_3^*| + |w_4^*|$$

Report your answer as $\lfloor \|\mathbf{w}^*\|_1 \rfloor$.

Q2: Finding the Optimal Pruning Strategy [3 pts]

You are given a binary classification task with **7 examples** and **7 features**. The weight vector $\mathbf{w} \in \mathbb{R}^7$ is already trained (note: at least one weight equals 0).

Dataset:

$$\mathbf{X} = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & -1 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 & -1 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 & 1 & -6 \\ 1 & 1 & 1 & 1 & 1 & 5 & -9 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

Task: You must prune **exactly two features**. This means:

1. Choose two feature indices $\{j, k\}$ where $j \neq k$ and $j, k \in \{1, 2, 3, 4, 5, 6, 7\}$.
2. Set $w_j = w_k = 0$ (equivalently, remove those columns from $\mathbf{X}$).
3. Compute the pruned score: $\mathbf{s}^{(j,k)} = \mathbf{X}\mathbf{w}^{(j,k)}$.

Output

Find the pair $\{j^*, k^*\}$ that maximizes the accuracy after pruning (without bias correction). Report the single integer $j^* \cdot k^*$ (the product of the two feature indices). Use 1-indexing, i.e., $j^*, k^* \in \{1, 2, 3, 4, 5, 6, 7\}$.

Q3: Bias Correction for Improved Performance [3 pts]

Use the best pruning pair $\{j^*, k^*\}$ from **Q2** and the corresponding pruned weight vector $\mathbf{w}^{(j^*,k^*)}$ (where $w_{j^*} = w_{k^*} = 0$ and other weights unchanged). The pruned score vector is $\mathbf{s}^{(j^*,k^*)} = \mathbf{X}\mathbf{w}^{(j^*,k^*)} \in \mathbb{R}^7$.
Task: You can now add an **integer bias term** b to every score:

$$\mathbf{s}_{\text{bias}}(b) = \mathbf{s}^{(j^*,k^*)} + b\mathbf{1},$$

where $\mathbf{1} \in \mathbb{R}^7$ is the all-ones vector and $b \in \{-3, -2, -1, 0, 1, 2, 3\}$. Predictions are made using the standard rule with threshold at 0.

1. Compute the accuracy for each allowed value of b.
2. Let b^* be a value that maximizes accuracy.
3. **Tie-breaking rule:** If multiple values achieve the same maximum accuracy, choose the b with smallest $|b|$ (closest to 0); if still tied, choose the more negative b.

Output

Report $\lfloor (j^* \cdot k^*) + |b^*| \rfloor$, where j^* and k^* are the feature indices from Q2. Use 1-indexing, i.e., $j^*, k^* \in \{1, 2, 3, 4, 5, 6, 7\}$.